

Byapari AI (B-AI) Project Report

1. Project Overview

Byapari AI is an AI-powered business intelligence platform for merchants who manage sales data in CSV format and need fast, practical guidance without building a traditional analytics stack. The system converts raw transaction data into business KPIs, trend analysis, forecasting, pricing suggestions, and AI-generated recommendations in English or Bengali. The project addresses a common merchant workflow: data exists, but insight does not. Its target users are small and growing merchants, especially Bangladeshi sellers working from spreadsheets and mobile-first operations. Key features include CSV upload and validation, KPI dashboards, business health scoring, sales forecasting, dynamic pricing guidance, AI insights grounded in merchant data, account history, and subscription-based usage limits.

2. Technology Stack

Frontend: Next.js 16, React 19, TypeScript, Tailwind CSS 4, Recharts.

Backend: FastAPI, Python 3.11, pandas, Prophet, Groq SDK.

Database: MongoDB via pymongo.

APIs: Firebase Auth, Stripe, Ollama, Google News RSS, open.er-api.com, ImgBB.

Deployment: Vercel frontend, Uvicorn-hosted backend.

3. System Architecture

The application uses a two-tier architecture. The Next.js frontend handles authentication, upload UX, dashboard rendering, pricing pages, and history/profile screens. The FastAPI backend performs data validation, analytics, forecasting, pricing heuristics, AI prompt assembly, and persistence. MongoDB stores user history and billing state, while external services enrich the response with market context and subscription handling.

4. Key Functionalities

CSV ingestion and validation, KPI and health scoring, Prophet-based forecasting, dynamic pricing suggestions, AI-generated insights, history and billing management, and Bengali/English localization.

5. Development & Deployment

The repository is split into backend and frontend applications. Backend routes live under `backend/app/routes`, while reusable logic resides in `backend/app/services`. Frontend API interactions are centralized and use context providers for authentication and language management.

6. Security & Performance Highlights

Authentication is handled with Firebase Auth. MongoDB stores history and billing data.

Performance improvements include lightweight retrieval for business tips, in-memory caching, and a stateless service design. Major cost centers are CSV processing, forecasting, and LLM inference.

7. Challenges & Future Improvements

Recommended enhancements include server-side Firebase token verification, automated testing and CI/CD pipelines, structured logging, unique upload paths, history search and pagination, and background processing for heavy analytics.

8. Conclusion

Byapari AI is a focused merchant intelligence platform that turns spreadsheet-based sales records into actionable business guidance using a lightweight architecture built on Next.js, FastAPI, MongoDB, and AI services.